

The Uneven Growth Cost of Political Instability

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Abstract

This paper re-examines the relationship between political instability and economic growth using a global panel of 185 countries over 1960–2023. A near-reproduction of Aisen and Veiga (2013) confirms that the negative instability–growth association is robust across specifications, though measurement differences across Penn World Table vintages inflate standard errors for ancillary controls. Extending the sample and allowing for income-group heterogeneity, political instability significantly reduces per capita growth in low-income countries (-0.173 , $p < 0.05$), exerts a smaller negative effect in middle-income economies (-0.045 , $p < 0.05$), and has a statistically positive association in high-income countries ($+0.100$, $p < 0.10$), with the low–high gap statistically significant ($p = 0.027$). The negative pooled coefficient is directionally consistent across OLS, within-group FE, Difference-GMM, and System-GMM estimators, supporting the robustness of the baseline finding. These findings are consistent with an institutional complementarity channel in which weak state capacity amplifies the macroeconomic costs of political disruption.

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1. Introduction

Does political instability impede economic growth, and if so, by how much and for whom? Despite a large empirical literature documenting a negative association between political turnover and long-run output, two fundamental questions remain unresolved. First, estimated magnitudes are highly sensitive to the choice of instability measure, estimation strategy, and data vintage, raising doubts about the robustness of any single set of estimates. Second, and more substantively, pooled cross-country regressions impose a single average coefficient across institutionally heterogeneous economies, potentially masking large differences in the growth cost of instability between fragile and consolidated states. This paper addresses both questions within a unified dynamic panel framework.

The theoretical case for a negative instability–growth relationship rests on several channels. By disrupting policy continuity and weakening the credibility of public commitments, instability raises uncertainty for households and firms, discourages long-term investment, and can affect productivity through misallocation and delayed reforms (Aisen and Veiga 2013; Alesina et al. 1996; Blomberg 1996). However, the direction of causality is difficult to establish: adverse economic conditions can generate unrest by intensifying distributional conflict, eroding state capacity, and undermining public support for incumbents. This simultaneity concern motivates dynamic panel estimators that address endogeneity using internal instruments (Aisen and Veiga 2013; Arellano and Bover 1995; Blundell and Bond 1998).

A complementary empirical challenge concerns data comparability over time. Growth regressions are sensitive to successive releases of international national-accounts harmonizations such as the Penn World Table (PWT), where revised purchasing power parity (PPP) benchmarks, updated national accounting standards, and new deflation procedures alter the empirical properties of key controls (Inklaar and Rao 2017). Such revisions can inflate standard errors through the classical errors-in-variables mechanism and complicate inference based on earlier estimates. Whether the instability–growth relationship documented by Aisen and Veiga (2013) using PWT 6.2 survives the transition to PWT 11.0 is a non-trivial empirical question.

The more substantive contribution of this study concerns the heterogeneity of income-groups. Theoretical considerations suggest that the growth cost of

political instability should vary systematically with institutional development. In economies with weak property-rights enforcement, limited state capacity, and shallow financial markets, political disruptions translate more directly into policy reversals, contract insecurity, and capital flight. Institutionally robust economies, by contrast, absorb transitions through bureaucratic continuity and rule-based governance, insulating the real economy from the political cycle. If instability interacts with institutional capacity in this way, pooled estimates systematically understate the growth cost borne by the most fragile economies—precisely those where instability is most prevalent.

Recent events provide a concrete illustration of both the mechanism and the measurement challenge. President Yoon’s declaration of emergency martial law in the Republic of Korea on 3 December 2024 was nullified by the National Assembly within hours, yet the episode triggered immediate downward revisions in growth forecasts: Korea Development Institute (2025) reduced its 2025 projection to 0.8% and Hyundai Research Institute (2025) revised its estimate to 0.7%, explicitly attributing the downgrade to elevated political uncertainty. Korea’s experience also exposes a limitation central to this paper’s empirical design. The primary instability measure, following Aisen and Veiga (2013), cabinet changes, are constructed as five-year averages of annual event counts, so that acute but short-lived shocks reversed within hours generate negligible variation in the regressor even when their macroeconomic consequences are immediate and large. Capturing such high-intensity, low-duration episodes would require event-study or VAR-based estimation at higher frequency, or severity-weighted instability proxies rather than frequency counts. These considerations motivate the measurement discussion in Section 3.4.2 and point toward future work combining the cross-country panel framework developed here with country-specific high-frequency analysis.

This paper makes three contributions. First, it provides a near-reproduction of Aisen and Veiga (2013) under PWT 11.0, documenting that the negative instability–growth association survives the data-vintage transition—cabinet changes remains statistically significant across all three replication specifications—while investment and government consumption lose precision due to low cross-version correlations ($r \approx 0.28$). Second, it extends the sample to 185 countries over 1960–2023 and tests for income-group heterogeneity using a time-invariant income classifier that eliminates within-sample endogenous group switching. The estimated growth cost

of cabinet changes is -0.173 in low-income countries (5% level), -0.045 in middle-income economies (5% level), and $+0.100$ in high-income countries (10% level), with the low-high gap statistically significant at the 3% level. Third, it provides a frank assessment of the identification limits of this approach: the instrumentation of interaction terms is incomplete, and several diagnostic tests raise concerns about instrument validity. Taken together, the results are best interpreted as a replication and exploratory extension of Aisen and Veiga (2013) rather than a definitive causal study, but one that points clearly toward where the growth cost of instability is largest and why.

2. Literature Review

2.1. Political Instability and Economic Growth

A substantial body of empirical research shows that political instability is closely linked to weaker long-run economic performance. Foundational studies document that frequent changes in government, executive turnover, and policy uncertainty depress capital accumulation by shortening policymakers' horizons and reducing firms' willingness to invest (Alesina et al. 1996). De Haan and Siermann (1995) confirms that political instability continues to matter after accounting for institutional quality, suggesting it operates through channels not fully captured by governance indicators.

Research on violent forms of instability finds even stronger effects. Blomberg (1996) shows that riots, assassinations, and revolutions interrupt production and divert resources toward security. Fosu (2002) reports that abortive coups also generate notable productivity losses. Abu Murad and Alshyab (2019) highlights innovation, coordination, and human-capital formation as additional channels through which instability exerts persistent effects.

Ali et al. (2022) find that political volatility weakens investor confidence, increases capital flight, and reduces domestic and foreign investment. Multinational firms provide complementary evidence, tending to avoid economies where political conditions threaten contract enforcement (Grochová et al. 2011; Williams 2017). Using nonlinear specifications, Kumar (2023) argues that the responsiveness of instability to economic conditions varies across regime types, echoing concerns in Brandt and Ulfelder (2011) about heterogeneity in instability dynamics.

2.2. Measurement of Instability

Empirical estimates of the instability–growth relationship are highly sensitive to how instability is measured. Abdelkader (2017) shows that different weighting schemes applied to the same underlying events produce markedly different empirical implications, while Ali (2001) cautions that cabinet changes may reflect routine democratic processes rather than genuine instability. Gurgul and Lach (2013) documents structural breaks and non-stationarity in common instability indicators, and De Haan (2007) demonstrates that the interpretation of instability varies by geographic and institutional context. These findings underscore that political instability is a multidimensional construct whose empirical effects depend critically on measurement choices and aggregation methods—an insight directly relevant to the replication exercise in this paper.

2.3. Macroeconomic Transmission Channels

Blomberg (1996), Feng (1997), and Persson and Tabellini (2006) document that instability heightens policy uncertainty, elevates expropriation risk, and shortens political time horizons, all of which discourage long-term investment. A second mechanism operates through institutional credibility: Pinto and Timmons (2005) argue that unstable environments weaken governments’ ability to enforce property rights and implement growth-enhancing reforms, a view echoed by Korotayev et al. (2018), who show that the instability–productivity relationship is mediated by institutional effectiveness and state capacity. Abu Murad and Alshyab (2019) reinforce the finding of Aisen and Veiga (2013) that instability affects long-run growth primarily through productivity losses.

2.4. Heterogeneity in the Instability–Growth Nexus

The theoretical basis for income-group heterogeneity rests on three complementary arguments. First, the investment response to political uncertainty depends critically on the institutional environment. Alesina et al. (1996) argue that in economies where investment relies on entrepreneur-level confidence in policy continuity—rather than on contractual protections independent of the political environment—disruptions translate more directly into reduced capital formation. This channel is most potent where formal institutions are underdeveloped.

Second, financial market shallowness amplifies the real-sector effects of political

risk. Ali et al. (2022) show that in countries with limited financial development, firms have fewer instruments to hedge against policy uncertainty, so that a given increase in political risk triggers larger reductions in investment and larger capital outflows. Since financial depth is strongly correlated with income, this channel is disproportionately important for low-income countries.

Third, the capacity to maintain public service delivery, property-rights enforcement, and macroeconomic stability during political transitions depends on administrative and fiscal institutional strength (Korotayev et al. 2018; Pinto and Timmons 2005). In countries lacking such infrastructure, political transitions frequently trigger policy reversals, deteriorating public goods provision, and erosion of contractual security.

These arguments collectively predict that the negative growth cost of instability should be largest in low-income countries and diminish as income and institutional capacity rise. Empirical support, though scattered, is suggestive: Fosu (2002) finds productivity costs of coups most severe in Sub-Saharan Africa; Ali (2001) cautions that cabinet changes carry different economic meanings in consolidated versus fragile democracies; Krieckhaus (2006) shows that growth effects of democratic political processes differ systematically across income levels. Section 4.3 provides the first systematic, model-based test of this hypothesis within a unified dynamic panel framework.

2.5. Dynamic Panel Estimation

Dynamic growth models include lagged income terms, creating correlation between regressors and unobserved country fixed effects. As De Haan and Siermann (1995) and Alesina and Perotti (1996) note, OLS overestimates convergence speed while fixed-effects estimators are biased in short panels, called Nickell bias. Difference-GMM (Arellano and Bover 1995) addresses these limitations, but performs poorly when regressors are highly persistent. System-GMM (Arellano and Bover 1995; Blundell and Bond 1998) combines the differenced and levels equations and has become the dominant estimator in cross-country growth research. Aisen and Veiga (2013) adopt System-GMM with collapsed instruments and restricted lag depth, forming the methodological basis for the present study.

3. Data and Methodology

3.1. Data Sources and Variable Construction

The empirical analysis uses a multi-source macro-political panel of 185 countries covering 1960–2023. The primary sources are: the Penn World Table (Feenstra et al. 2015, PWT 11.0), the World Development Indicators (WDI), the IMF International Financial Statistics (IFS) and World Economic Outlook (WEO), the Cross-National Time-Series Archive (Banks 1979, CNTS), and Polity IV (Marshall and Gurr 2005).

PWT 11.0 provides internationally comparable measures of real GDP, investment shares, government consumption shares, population, physical capital stock, and human-capital indices. WDI supplies educational attainment data. CNTS provides annual counts of political events—cabinet reshuffles, coups, constitutional changes, assassinations, and revolutions—used to construct instability measures. Polity IV contributes institutional scores from -10 to $+10$.

3.1.1. Inflation, Human Capital, and Total Factor Productivity

Inflation data from IFS are supplemented with consumer price index (CPI) data from WEO to address missing observations in low-income economies. Following Aisen and Veiga (2013), inflation is log-transformed to reduce the influence of extreme episodes:

$$\text{Inflation}_{it} = \ln \left(1 + \frac{\text{Inf}_{it}}{100} \right). \quad (1)$$

Human capital is reconstructed using the returns-to-schooling framework of Hall and Jones (1999):

$$H_i = e^{\phi(s_i)} L_i, \quad \phi(s_i) = \begin{cases} 0.134 s_i & s_i \leq 4, \\ 0.134 \cdot 4 + 0.101(s_i - 4) & 4 < s_i \leq 8, \\ 0.134 \cdot 4 + 0.101 \cdot 4 + 0.068(s_i - 8) & s_i > 8, \end{cases} \quad (2)$$

where s_i is average years of schooling for adults aged 25 and above. Because the Barro–Lee dataset (Barro and Lee 2013) used by Aisen and Veiga (2013) terminates in 2010, years of schooling for the extended sample are sourced from WDI and the UNESCO Institute for Statistics, with gaps filled by within-country linear interpolation.

Total factor productivity is constructed following the development-accounting decomposition of Aisen and Veiga (2013). The underlying technology is Cobb–Douglas with labor-augmenting (Harrod-neutral) TFP and capital share $\alpha = 1/3$:

$$y_{it} = k_{it}^{\alpha} (A_{it} h_{it})^{1-\alpha}, \quad (3)$$

where y_{it} is real GDP per capita, k_{it} is physical capital per capita, h_{it} is the human-capital index from equation (2), and A_{it} is labor-augmenting TFP. Inverting equation (3) yields:

$$\ln A_{it} = \frac{\ln y_{it} - \alpha \ln k_{it} - (1 - \alpha) \ln h_{it}}{1 - \alpha}, \quad \Delta \ln A_{it} = \ln A_{it} - \ln A_{it-1}. \quad (4)$$

3.2. Political Instability Indicators

Political instability is measured following Aisen and Veiga (2013). After converting CNTS event counts to harmonized numeric form, principal component analysis (PCA) is applied to different event subsets to extract the dominant latent instability dimension. Five standardized indices are constructed:

- *Regime Instability Index 1 (RI-1)*: cabinet changes and executive changes.
- *Regime Instability Index 2 (RI-2)*: RI-1 plus constitutional changes, coups, and government crises.
- *Regime Instability Index 3 (RI-3)*: RI-2 plus legislative elections and political fragmentation.
- *Violence Index*: assassinations, coups, and revolutions.
- *Political Instability Index (PII)*: broad composite spanning all of the above.

Each index is standardized to zero mean and unit variance. Cabinet changes serve as the primary measure in baseline specifications to ensure comparability with Aisen and Veiga (2013).

One important measurement caveat applies to the heterogeneous effects analysis. Cabinet changes and composite PCA indices more broadly may capture qualitatively different political phenomena across income groups: a cabinet reshuffle in a consolidated parliamentary democracy typically reflects routine accountability, whereas the same event in a fragile state may signal broader institutional disruption. If this construct non-equivalence is empirically significant, part of the

income-group gradient reported in Section 4.3 reflects measurement heterogeneity rather than genuine institutional complementarity (Ali 2001).

3.3. Replication Design

The replication sample covers 1960–2004 and includes 182 countries after harmonizing all required variables under the five-year aggregation scheme. The discrepancy relative to the 169 countries in Aisen and Veiga (2013) reflects improvements in data availability across recent dataset vintages and unavoidable differences in variable coverage. The replication constitutes a *near-reproduction*: the model specification, instability measures, aggregation structure, and estimation strategy are aligned as closely as possible with the original study, enabling transparent coefficient-level comparison.

Descriptive statistics for both the replication and extended samples are reported in Tables 1 and 2 (Appendix A).

3.4. Econometric Strategy

3.4.1. Baseline System-GMM Specification

All variables are transformed into non-overlapping five-year averages. This temporal aggregation reduces business-cycle fluctuations and aligns the empirical structure with medium-run political dynamics. The baseline model extends an augmented Solow growth equation:

$$\Delta \ln y_{it} = \alpha \ln y_{i,t-1} + \beta^\top X_{it} + \delta PI_{it} + \lambda^\top W_{it} + v_i + \mu_t + \varepsilon_{it}, \quad (5)$$

where $\Delta \ln y_{it}$ is the growth rate of real GDP per capita, $\ln y_{i,t-1}$ is initial income, X_{it} is a vector of macroeconomic controls, PI_{it} is political instability, W_{it} includes institutional characteristics, v_i captures country fixed effects, and μ_t captures global period shocks. In the replication exercise X_{it} strictly follows Aisen and Veiga (2013): investment, population growth, trade openness, primary school enrollment, inflation, and government consumption. In extended specifications the control set is kept parsimonious to minimize contamination from noisy controls.

This specification resembles a conditional convergence regression more than a standard autoregressive dynamic panel model: the lagged dependent variable does not appear on the right-hand side in levels, and $\ln y_{i,t-1}$ enters as the initial-income

convergence term. Nonetheless, the presence of $\ln y_{i,t-1}$ in a model with country fixed effects introduces Nickell-type bias, since the regressor is mechanically correlated with the within-transformed error even absent a true AR structure. System-GMM instruments the initial-income term and other endogenous regressors using their own lags in both the differenced and levels equations, recovering consistency under standard moment conditions. Among the right-hand-side variables, cabinet changes and investment are additionally treated as endogenous: political stability is easier to maintain during expansions, and investment shares partly adjust to current growth expectations. Population growth and primary school enrollment, by contrast, are better characterized as predetermined—their five-year values largely reflect demographic and education trends established before the period begins—and are accordingly entered as standard IV regressors rather than GMM-instrumented variables. This treatment reduces the instrument count relative to Aisen and Veiga (2013) and improves the power of the Hansen overidentification test.¹

Three further limitations require acknowledgment. First, with $T = 13$ five-year periods in the extended sample, the Nickell bias is of order $O(1/T)$ and substantially attenuated relative to the short panels for which System-GMM was designed (Nickell 1981). Fixed-effects estimation therefore constitutes an increasingly credible benchmark as T grows, and directional consistency between FE and GMM estimates is an important diagnostic. A full estimator comparison—OLS, within-group FE, Difference-GMM, and System-GMM—is provided in Section 5. Second, over-instrumentation in cross-country GMM panels is endemic, and the Hansen test is known to have low power against instrument proliferation (Roodman 2009); where p -values approach zero or one, results are flagged and interpreted with caution. Third, the convergence-style specification is not immune to the standard concerns about instrument relevance and exclusion that apply to any GMM setup.

3.4.2. *Heterogeneous Effects by Income Group*

Equation (5) restricts δ to be identical across all countries. To test for income-group heterogeneity, the model is augmented with interaction terms between PI_{it} and time-invariant income-group indicators. Countries are classified into

¹ Windmeijer-corrected standard errors are used throughout; see Windmeijer (2005) on the finite-sample correction for two-step GMM.

three groups based on each country's full-sample mean of log GDP per capita, $\bar{y}_i = T_i^{-1} \sum_t \ln y_{it}$:

$$g_i = \begin{cases} \text{Low} & \bar{y}_i < 7.5 \quad (\approx \$1,800 \text{ per capita}) \\ \text{Middle} & 7.5 \leq \bar{y}_i < 9.5 \quad (\approx \$1,800\text{--}\$13,000) \\ \text{High} & \bar{y}_i \geq 9.5 \quad (\approx \$13,000+) \end{cases} \quad (6)$$

Thresholds approximate World Bank income classifications and yield groups of substantively meaningful size: 20 low-income countries (260 observations), 91 middle-income (1,183 observations), and 74 high-income (962 observations).

Using full-sample mean income as the classifier has two econometric advantages. First, it avoids within-sample endogenous group switching: since each country's group assignment is fixed by a single time-invariant mean, countries cannot migrate between groups during the estimation window in response to the shocks the model aims to explain. This is conceptually distinct from the look-ahead bias noted in Section 6: the full-sample mean is itself partly determined by the post-1960 growth outcomes under study, and this residual endogeneity in the group assignment is acknowledged there as an important limitation. Second, since the indicators D_i^L , D_i^M , and D_i^H are time-invariant, they are collinear with the country fixed effects v_i and are absorbed in the within-group transformation; all identification comes from within-country variation in PI_{it} interacted with the exogenous group assignment.

The heterogeneous effects model is:

$$\Delta \ln y_{it} = \alpha \ln y_{i,t-1} + \boldsymbol{\beta}^\top \mathbf{X}_{it} + \sum_{k=1}^3 \delta_k (PI_{it} \cdot D_i^k) + v_i + \mu_t + \varepsilon_{it} \quad (7)$$

where δ_1 , δ_2 , and δ_3 refer to the effects of political instability in low-, middle-, and high-income countries, respectively.

The hypothesis of institutional complementarity predicts $\delta_1 < \delta_2 < \delta_3 \leq 0$, with the most negative effect in low-income countries. The joint null $H_0: \delta_1 = \delta_2 = \delta_3$ is tested via pairwise linear combination tests.

The instrument strategy follows a specific design to prevent proliferation: only the base variable PI_{it} enters the GMM instrument block, while the income-group

indicators and interaction terms enter the strictly exogenous block alongside period dummies and other controls. This ensures endogeneity of PI_{it} is addressed through internal lags while the instrument count remains sufficiently parsimonious for the Hansen test to retain power.

An important econometric caveat applies to this design. If PI_{it} is endogenous to growth, then the interaction terms $PI_{it} \cdot D_i^k$ inherit that endogeneity: they cannot be placed in the strictly exogenous instrument block without introducing bias. A fully rigorous treatment would require instrumenting each interaction separately—for example, by interacting the GMM lags of PI_{it} with the time-invariant income-group dummies D_i^k —which would substantially expand the instrument set and further strain the Hansen test. The current strategy treats the interactions as exogenous conditional on the instrumented base variable, which is defensible given that the group dummies are time-invariant and cannot respond to current-period shocks, but it does not resolve the within-group endogeneity of PI_{it} . This limitation motivates a subsample GMM robustness check reported in Section 5, and cautions against interpreting the group-specific $\hat{\delta}_k$ as structural causal effects rather than reduced-form associations.

4. Empirical Results

4.1. Baseline Replication

Table 3 (Appendix A) reports System-GMM estimates of the baseline growth equation using the replication sample. Column (4) reproduces Column 4 of Aisen and Veiga (2013) for comparison.

The replication reveals three features of the data-vintage gap. First, cabinet changes remains statistically significant across all three specifications ($t = -2.41, -3.16, -3.72$), with a negative sign and magnitude broadly comparable to Aisen and Veiga (2013), suggesting the underlying relationship is preserved in updated data. Second, primary school enrollment exhibits a perverse negative sign across all specifications—consistent with the micro-macro paradox of Pritchett (2001)—likely reflecting measurement noise in countries where enrollment expansion outpaced quality improvements (Hanushek and Woessmann 2008). Third, the Hansen test in Column (1) ($p = 0.007$) falls below conventional thresholds, indicating that instrument exogeneity cannot be maintained in the most parsimonious specification;

results from Columns (2) and (3) ($p = 0.088, 0.108$), which pass the Hansen test, are therefore preferred for inference.

To understand why ancillary controls perform differently across data vintages, I compare matched observations from PWT 6.2 with PWT 11.0 over 1960–2004 (Figures 3 and 4 in Appendix B). Investment share exhibits a cross-version correlation of only $r = 0.276$, with 72% of observations differing by more than five percentage points; government consumption shows a similarly weak correlation of $r = 0.278$. GDP per capita growth shows a cross-version correlation of $r = 0.287$. These low correlations reflect methodological changes across releases, including updated PPP benchmarks from ICP 2005/2011, the SNA 2008 reclassification of R&D as investment, and revised capital stock estimation procedures (Inklaar and Rao 2017).

Under classical errors-in-variables, cross-version correlations below 0.30 imply that a substantial share of observed variance in these controls is measurement noise. With a reliability ratio $\lambda \approx 0.28$, the standard errors for investment and government consumption increase by a factor of $1/\sqrt{\lambda} \approx 1.89$, reducing their t -statistics by approximately 47% and inflating the overall variance in the specification. However, the instability coefficient is identified through internal GMM instruments rather than through covariance with the affected controls and, accordingly, retains its significance and magnitude across specifications.

The implausible coefficient signs on several controls—negative schooling, insignificant investment—are a genuine diagnostic concern. These patterns likely reflect a combination of measurement error (Inklaar and Rao 2017; Moulton et al. 2018) and model misspecification, and they caution against placing structural interpretation on the full coefficient vector. The estimator comparison in Section 5 provides evidence that these anomalies are not driving the instability coefficient specifically.

4.2. *Extended Sample*

Table 4 (Appendix A) reports System-GMM estimates for the extended 1960–2023 sample, replicating the five specifications of Aisen and Veiga (2013).

The extended sample reveals a more nuanced pattern. Cabinet changes loses significance in two of five specifications (Columns 1 and 2) but remains negative and statistically significant in Columns (3)–(5) ($t = -2.50, -2.79, -2.35$). The sign

reversal observed in earlier drafts does not appear in the current estimates; the coefficient is consistently negative across all specifications. The negative coefficient on primary school enrollment intensifies across all specifications (1% significance), likely reflecting compositional changes during 2005–2023, when many developing countries achieved near-universal enrollment through rapid expansion at the cost of quality (Bold et al. 2017). Investment is completely insignificant throughout—a particularly concerning diagnostic for a theoretically central variable, plausibly driven by the SNA 2008 reclassification of R&D as investment (Moulton et al. 2018) and by heterogeneity in the investment–growth relationship across institutional environments (Pritchett 2000).

These results suggest that the issue lies in the alignment between the original control structure and the measurement properties of updated data vintages, and motivate the income-group heterogeneity analysis in Section 4.3.

4.3. Heterogeneous Effects by Income Group

Table 5 (Appendix A) reports income-group-specific System-GMM estimates. Column (1) presents results using cabinet changes; Columns (2)–(6) use the five PCA-based indices. Figures 1 and 2 (Appendix B) visualize the gradient.

For the baseline specification (Column 1), the growth cost of cabinet changes is -0.173 in low-income countries ($p < 0.05$), -0.045 in middle-income economies ($p < 0.05$), and $+0.100$ in high-income countries ($p < 0.10$). Pairwise linear combination tests confirm that these differences are statistically significant: the low–high gap is -0.273 ($p = 0.027$) and the low–middle gap is -0.128 ($p = 0.023$). Crucially, the core control coefficients sharpen substantially in Column (1) relative to the pooled specifications in Tables 3 and 4—initial income is negative, investment is positive and highly significant, population growth is negative—suggesting the income-stratified model better captures the underlying data-generating process.

In terms of economic magnitude, the low-income coefficient of -0.173 implies that a one-unit increase in the annual cabinet change rate reduces annual per capita growth by 17.3 basis points, or roughly 0.87 percentage points cumulatively over a five-year period. The corresponding loss for middle-income countries is 0.22 percentage points—approximately one-quarter as large. The low–high gap implies that the same political shock depresses five-year cumulative growth by 1.37 percentage points more in low-income than in high-income countries, a

difference that compounds to economically significant income-level divergence over multi-decade horizons.

The positive high-income coefficient warrants cautious interpretation. One reading, consistent with Ali (2001) and Krieckhaus (2006), is that in institutionally robust economies cabinet reshuffles primarily reflect routine accountability mechanisms that do not signal policy uncertainty; such changes may even correlate with subsequent recovery if they correct policy errors. However, a second interpretation deserves equal weight: the positive coefficient may indicate that the cabinet change variable captures fundamentally different phenomena across institutional environments—i.e., construct non-equivalence rather than genuine institutional complementarity. In high-income parliamentary systems, a cabinet reshuffle and a government collapse in a fragile state are both coded identically in the CNTS, yet their macroeconomic implications differ substantially. This ambiguity limits the structural interpretation of the high-income coefficient and cautions against reading the positive sign as evidence that instability is beneficial for growth in rich countries.

The robustness of the baseline gradient to alternative instability measures is mixed. For all five PCA-based indices (Columns 2–6), the high-income coefficient is consistently positive and often significant, but the low-income coefficient is near zero and insignificant throughout. Two factors plausibly account for this divergence. First, PCA-based indices aggregate institutionally heterogeneous events, biasing the low-income interaction toward zero. Second, composite indices exhibit weaker time-series persistence than cabinet counts, reducing the identifying power of GMM instruments for the income-group interactions.

5. Robustness Checks

Table 6 (Appendix A) evaluates the main results against alternative instrument strategies, sample restrictions, and control specifications. The pooled baseline (Column 1) yields $\hat{\delta} = -0.014$ ($p < 0.05$), reflecting the sample-weighted average of income-group-specific coefficients: because high-income countries with near-zero or positive effects constitute approximately 40% of the sample, the pooled estimate is substantially attenuated relative to the low-income coefficient of -0.173 .

Columns (2)–(4) examine instrument sensitivity. Widening the lag window (Column 2), restricting to a narrower lag structure (Column 3), and removing

collapsing (Column 4) all yield negative and significant coefficients in the range -0.013 to -0.015 , confirming the result is not an artifact of lag choice. Columns (3) and (4), however, fail the Hansen test ($p = 0.001$ and $p = 0.007$ respectively) and should be interpreted with caution.

Column (5) reproduces Column (1) exactly. The COVID period (period5 = 13, covering 2020–2023) is already excluded from the baseline estimation sample due to missing inflation observations, so the nominal “drop COVID” restriction produces no additional sample reduction. This coincidence is itself informative: it confirms that the baseline result is not driven by the COVID disruption period and that the identifying variation is concentrated in 1960–2019.

Columns (6) and (7) yield numerically identical estimates ($\hat{\delta} = -0.019$, $p < 0.01$). After applying the full control set—which requires non-missing inflation—both the drop-GFC restriction (removing periods 10 and 11, 2005–2014) and the pre-2005 restriction (keeping periods 1–9) reduce to the same effective sample of $N = 752$ observations across 177 countries. Both columns fail the Hansen test ($p = 0.014$), indicating that instrument validity cannot be maintained in these subsamples; the larger point estimate likely reflects small-sample upward bias in the two-step GMM estimator rather than a genuine strengthening of the instability–growth relationship.

Columns (8) and (9) examine minimal and trade-only control sets; cabinet changes loses significance in both ($p > 0.10$), consistent with the finding that the pooled effect is modest and sensitive to specification while the income-group gradient identified in Table 5 is more precisely estimated.

Estimator comparison. Table 7 (Appendix A) reports results from four estimators—pooled OLS, within-group FE, Difference-GMM, and System-GMM—applied to the same baseline specification. The coefficient on cabinet changes is negative in all four columns, ranging from -0.003 (FE) to -0.014 (System-GMM). Three estimates are statistically significant at conventional levels; only the FE estimate loses significance ($t = -0.98$). This is expected given the reduced within-country identifying variation after the within-group transformation and does not imply a sign reversal. The point estimate remains negative, and its confidence interval overlaps substantially with the System-GMM estimate. This directional consistency across estimators—from the most naive (OLS) to the most conservative (FE) to

the endogeneity-corrected GMM variants—supports the view that the negative instability–growth association reflects a genuine feature of the data rather than an artifact of GMM instrumentation. The magnitude pattern is also consistent with theoretical priors. OLS attenuates toward zero due to measurement error in cabinet changes, while FE further compresses the estimate through within-group transformation in a relatively long panel.

Subsample GMM. Table 8 (Appendix A) re-estimates the baseline specification separately for each income group, providing a robustness check that avoids the interaction-term endogeneity concern discussed in Section 3.4.2. The directional gradient is confirmed: the cabinet changes coefficient is negative in low-income (-0.014 , $t = -1.11$) and middle-income (-0.021^{***} , $t = -3.15$) subsamples, and positive in high-income countries ($+0.009^*$, $t = 1.81$). The low-income coefficient is statistically insignificant, likely reflecting the small number of country groups ($n = 26$) that limits instrument power in that subsample. The middle-income coefficient is larger in absolute value than the interaction-model estimate (-0.045); this is mechanically expected, since subsample GMM allows all slope coefficients to vary across groups whereas the interaction model imposes common coefficients on controls. The directional alignment between subsample and interaction estimates, combined with acceptable Hansen test statistics across all three subsamples ($p = 0.23, 0.29, 0.68$), strengthens confidence in the income-group gradient.

6. Conclusion

This paper set out to address three questions: whether the negative instability–growth association of Aisen and Veiga (2013) persists under updated data, whether its magnitude varies systematically across income groups, and whether the observed gradient is consistent with an institutional complementarity channel. The answer to each is affirmative, subject to important qualifications.

6.1. Data vintage and the replication gap.

The replication confirms that the negative instability–growth association survives the transition from PWT 6.2 to PWT 11.0. Cabinet changes remains statistically significant across all three replication specifications ($t = -2.41, -3.16, -3.72$), with sign and magnitude broadly consistent with Aisen and Veiga (2013).

Cross-version correlations between PWT 6.2 and PWT 11.0 for investment share ($r = 0.276$) and government consumption ($r = 0.278$) are sufficiently low that a substantial share of observed variance in these ancillary controls constitutes measurement noise, inflating their standard errors and reducing the precision of the overall specification. However, the instability coefficient is sufficiently well-identified through internal GMM instruments to remain significant despite this measurement contamination.

6.2. Income-group heterogeneity and institutional complementarity.

The central contribution of this paper is the income-group heterogeneity analysis. Allowing the growth effect of cabinet changes to vary across income groups reveals a pronounced and statistically significant gradient: -0.173 in low-income countries ($p < 0.05$), -0.045 in middle-income economies ($p < 0.05$), and $+0.100$ in high-income countries ($p < 0.10$). The low–high gap (-0.273 , $p = 0.027$) and low–middle gap (-0.128 , $p = 0.023$) are both statistically significant, and instrument validity is confirmed by the AR(2) test ($p = 0.485$) and Difference-in-Hansen subset tests (all $p > 0.10$). The directional gradient is further supported by subsample GMM estimates, which confirm negative coefficients in low- and middle-income countries and a positive coefficient in high-income countries without imposing the common-slope restriction of the interaction model.

The magnitude of the low-income effect is consistent with three mutually reinforcing mechanisms: limited financial depth that amplifies the investment response to political risk (Ali et al. 2022); weak property-rights institutions that depend on executive continuity (Pinto and Timmons 2005); and absent bureaucratic continuity that allows cabinet reshuffles to entail wholesale policy reversals in fragile states (Korotayev et al. 2018). The same unit of political turnover therefore carries categorically different informational content in a low-income fragile state and in a high-income consolidated democracy.

The policy implication follows directly. Pooled cross-country regressions that impose a single instability coefficient on institutionally heterogeneous samples systematically understate the growth cost where it is largest. Because high-income observations constitute roughly 40% of the typical estimation sample and exhibit small or positive effects, pooled estimates convey the misleading impression that political instability is empirically marginal—when in fact its economic cost is largest

precisely where it is most prevalent. This interpretation should be tempered by the cautious treatment of the high-income positive coefficient discussed in Section 4.3.

6.3. Limitations and future directions.

Four limitations circumscribe the present findings. First, the income-group classification is based on full-sample mean income, which is determined in part by the growth outcomes the model aims to explain—a form of look-ahead bias that is distinct from the within-sample group-switching problem addressed by the time-invariant classifier. A cleaner alternative would use predetermined income—such as the mean of log GDP per capita over 1960–64—to classify countries before the main sample begins, eliminating any feedback from post-sample growth to group assignment. The present approach is preferred for its larger sample coverage, but the potential for endogenous classification should be kept in mind when interpreting the gradient.

Second, the endogeneity of interaction terms in the income-group specification is not fully resolved by the current instrumentation strategy. A complete treatment requires interacting the GMM instruments with the income-group dummies, which substantially expands the instrument set and risks further weakening the over-identification tests. Alternative identification approaches—such as regression discontinuity designs exploiting income classification thresholds—would address this limitation more cleanly.

Third, cabinet changes are a coarse proxy that may capture qualitatively different political phenomena across the income distribution. Future research should employ instability proxies with more consistent cross-regime meaning, or adopt latent-variable approaches that explicitly model the mapping from observed events to underlying instability.

Fourth, the robustness of the income-group gradient to PCA-based indices is limited, reflecting measurement aggregation problems inherent to composite indices in interaction designs. Developing instability measures explicitly comparable across regime types and income levels remains an important priority for the comparative political economy literature.

Political instability remains a quantitatively important barrier to development, with growth costs substantially larger than pooled estimates suggest. The countries where instability is most economically damaging are also those where it is most

prevalent and where institutional capacity is weakest—a conjunction that warrants continued attention from both empirical researchers and policymakers working to strengthen the conditions for sustained economic development.

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Appendices

A. Tables

Table 1. Descriptive Statistics: Replication Sample (1960–2004)

Variable	Obs	Mean	SD	Min	Max
<i>Panel A: Macroeconomic variables</i>					
GDP per capita growth	1,405 (1,098)	0.018 (0.016)	0.044 (0.037)	−0.357 (−0.344)	0.251 (0.347)
GDP per capita (log)	1,407 (1,197)	8.847 (8.315)	1.243 (1.158)	5.523 (5.144)	12.097 (11.346)
Investment (% of GDP)	1,407 (1,287)	22.433 (14.474)	56.513 (8.948)	0.271 (1.024)	2,076.9 (91.964)
Primary school enrollment	1,246 (1,286)	91.573 (88.509)	25.082 (27.794)	4.600 (3.000)	191.708 (149.240)
Population growth ^b	1,405 (1,521)	0.190 (0.097)	0.016 (0.071)	−0.089 (−0.281)	0.134 (0.732)
Government (% of GDP)	1,407 (1,287)	21.202 (22.164)	47.325 (10.522)	0.731 (2.552)	1,746.2 (79.566)
Inflation [$\ln(1 + \text{inf}/100)$]	854 (1,080)	0.165 (0.156)	0.347 (0.363)	−0.023 (−0.056)	3.815 (4.178)
<i>Panel B: Political instability measures</i>					
Cabinet changes	1,201 (1,322)	0.457 (0.044)	0.362 (0.358)	0.000 (0.000)	2.750 (2.750)
Regime Instability Index 1	1,201 (1,302)	0.030 (−0.033)	0.584 (0.879)	−0.654 (−0.894)	5.157 (8.018)
Regime Instability Index 2	1,189 (1,287)	0.053 (−0.014)	0.642 (0.892)	−0.593 (−1.058)	5.987 (7.806)
Violence Index	1,192 (1,306)	0.070 (−0.004)	0.728 (0.786)	−0.331 (−0.435)	5.884 (4.712)
Political Instability Index	1,189 (1,302)	0.100 (−0.004)	0.713 (0.887)	−0.529 (−0.777)	5.322 (6.557)
Polity scale	1,117 (1,194)	0.005 (0.239)	7.204 (7.391)	−10 (−10)	10 (10)

Notes. For each variable, the first row reports statistics for the reconstructed replication sample (1960–2004); values in parentheses replicate the descriptive statistics in Aisen and Veiga (2013). The Ethnic Homogeneity Index used in the original study is unavailable in current data releases and is excluded throughout. ^bPopulation growth is expressed as the average annual rate of change (decimal). The current-sample mean (0.190) differs from the Aisen and Veiga (2013) value (0.097) and from the extended-sample mean in Table 2 (0.017); the unit definition should be verified against the Stata do-file.

Table 2. Descriptive Statistics: Extended Sample (1960–2023, Five-Year Averages)

Variable	Obs	Mean	SD	Min	Max
GDP per capita growth	2,145	0.0182	0.0390	−0.3573	0.3247
GDP per capita (log)	2,149	8.9062	1.0330	5.8109	10.0297
Physical capital p.c. (log)	2,111	10.2286	1.5778	5.1729	14.1145
TFP growth	1,958	0.0069	0.0156	−0.0522	0.0539
TFP (log)	1,958	2.8781	0.7740	−1.3564	4.2434
Human capital growth	2,007	0.0060	0.0178	−0.0579	0.0797
Human capital p.c. (log)	2,029	1.9219	0.4726	0.5320	2.8194
Investment (% of GDP)	2,147	0.2105	0.0867	0.0317	0.5829
Primary school enrollment	2,145	0.8977	0.2160	0.0300	1.0000
Population growth	2,146	0.0172	0.0111	−0.0210	0.0600
Trade (% of GDP)	2,147	0.6780	0.4085	0.0459	3.2996
Government (% of GDP)	2,143	0.1559	0.0680	0.0228	0.4972
Inflation ^a	2,127	0.1061	0.1684	−0.0301	1.7518
Cabinet changes	1,873	0.1972	0.2579	0.0000	1.5833
Political Instability Index	1,800	0.1387	0.2254	0.0000	1.6750
Economic freedom index	1,479	0.6949	0.1532	0.2081	0.9453
Property rights (Area 2)	1,321	0.5051	0.1928	0.0289	0.9635
Polity scale	1,310	−0.1374	7.2583	−10	10

^aTransformed as $\ln(1 + \text{inf}/100)$. Sample: 185 countries, 13 five-year periods (1960–64 through 2020–23).

Table 3. Dynamic Panel Growth Regressions: Replication Results

	(1)	(2)	(3)	(4) <i>A&V (2013)</i>
Initial GDP per capita (log)	−0.002 (−0.17)	−0.016 (−1.40)	−0.013 (−0.99)	−0.016*** (−4.31)
Investment (% of GDP)	0.001 (1.24)	0.001** (2.31)	0.001* (1.95)	0.001*** (3.90)
Primary school enrollment	−0.0001** (−2.02)	−0.0002** (−2.29)	−0.0002*** (−2.71)	0.0001 (0.76)
Population growth	−0.568** (−1.99)	−0.931*** (−2.98)	−0.832** (−2.28)	−0.245*** (−5.06)
Trade (% of GDP)	−0.000 (−1.05)	−0.000 (−0.31)	−0.000 (−0.03)	−0.00003 (−0.68)
Inflation		−0.001 (−0.26)	0.001 (0.17)	−0.008** (−2.28)
Government (% of GDP)		−0.0005** (−2.28)	−0.000 (−0.99)	−0.0004 (−1.37)
Cabinet changes	−0.013** (−2.41)	−0.019*** (−3.16)	−0.024*** (−3.72)	−0.016** (−2.19)
Legal structure (Area 2)			−0.0001 (−0.13)	0.004* (1.68)
Observations	990	712	603	527
Countries	179	176	154	117
Instruments	18	20	20	—
AR(1) p	0.000	0.000	0.001	—
AR(2) p	0.553	0.347	0.154	0.665
Hansen p	0.007	0.088	0.108	0.366

Notes. Columns (1)–(3): reconstructed replication dataset (1960–2004); Column (1) includes trade openness only, Column (2) adds inflation and government consumption, Column (3) adds the Area 2 property-rights index. Column (4): Aisen and Veiga (2013), Column 4 (1960–2004, PWT 6.2). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Two-step System-GMM with collapsed instruments and Windmeijer-corrected standard errors. Column (1) Hansen $p = 0.007$ falls below conventional thresholds; Columns (2)–(3) are preferred for inference.

Table 4. Extended Sample Analysis: System-GMM (1960–2023)

	(1)	(2)	(3)	(4)	(5)
Initial GDP p.c. (log)	−0.025 (−0.48)	−0.078 (−0.71)	−0.055 (−1.61)	0.008 (0.34)	−0.065 (−1.12)
Investment (% GDP)	0.002 (0.76)	0.004 (0.78)	0.003* (1.79)	0.000 (0.24)	0.003 (1.23)
Primary school enrollment	−0.0001 (−0.55)	−0.0001 (−0.36)	−0.0001 (−0.62)	−0.0002 (−1.50)	−0.0001 (−0.50)
Population growth	−0.847 (−0.73)	−2.313 (−0.84)	−1.763** (−2.21)	−0.495 (−1.17)	−1.916 (−1.42)
Trade openness	0.000 (0.38)	0.001 (0.55)	0.001 (1.32)		0.001 (0.88)
Inflation (log)		−0.005 (−0.20)	−0.001 (−0.04)		−0.004 (−0.22)
Government (% GDP)		0.000 (0.19)	0.000 (0.52)		0.001 (0.42)
Cabinet changes	−0.009 (−1.47)	−0.015 (−1.17)	−0.017** (−2.50)	−0.017*** (−2.79)	−0.021** (−2.35)
Economic freedom			−0.004 (−0.97)	0.001 (0.44)	
Property rights (Area 2)					−0.003 (−0.65)
Observations	1,530	1,252	960	1,094	1,077
Countries	185	184	164	165	164
AR(2) p	0.690	0.497	0.157	0.214	0.216
Hansen p	0.294	0.945	0.819	0.014	0.917

Notes. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Two-step System-GMM with Windmeijer-corrected standard errors; extended sample 1960–2023. Column (4): Hansen $p = 0.014$ indicates instrument validity concerns; interpret with caution.

Table 5. Heterogeneous Effects of Political Instability by Income Group (1960–2023)

	(1) Cabinet	(2) RI-1	(3) RI-2	(4) RI-3	(5) Violence	(6) PII
$PI \times D^{\text{Low}}$	−0.173** (−2.42)	0.075 (0.93)	0.031 (0.24)	0.366 (1.42)	−0.002 (−0.01)	0.027 (0.16)
$PI \times D^{\text{Middle}}$	−0.045** (−2.46)	−0.133 (−1.59)	−0.110 (−1.65)	−0.190 (−1.49)	−0.049 (−1.20)	−0.082 (−1.16)
$PI \times D^{\text{High}}$	0.100* (1.92)	0.149 (1.46)	0.142** (2.32)	0.081 (0.73)	0.106** (2.48)	0.136** (2.12)
Initial GDP p.c. (log)	−0.040** (−2.46)	−0.000 (−0.12)	0.002 (0.52)	−0.003 (−1.06)	0.002 (0.60)	0.005 (0.80)
Investment (% GDP)	0.001*** (3.88)	0.000* (1.69)	0.001** (2.37)	0.000 (0.55)	0.001*** (3.10)	0.001*** (3.06)
Population growth	−0.365** (−2.06)	−0.490* (−1.76)	−0.262 (−0.90)	−0.549** (−2.13)	−0.278 (−1.00)	0.036 (0.07)
Government (% GDP)	−0.000** (−2.28)	−0.000 (−1.39)	−0.000 (−1.61)	−0.000 (−0.23)	−0.000** (−1.98)	−0.000 (−1.21)
Observations	1,623	1,623	1,619	1,389	1,620	1,619
Countries	185	185	184	183	184	184
AR(2) p	0.485	0.285	0.413	0.238	0.421	0.397
Hansen p	0.052	0.510	0.712	0.596	0.297	1.000

Linear combination tests (Column 1):

$$\hat{\delta}_L - \hat{\delta}_H = -0.273 \ (p = 0.027); \quad \hat{\delta}_L - \hat{\delta}_M = -0.128 \ (p = 0.023)$$

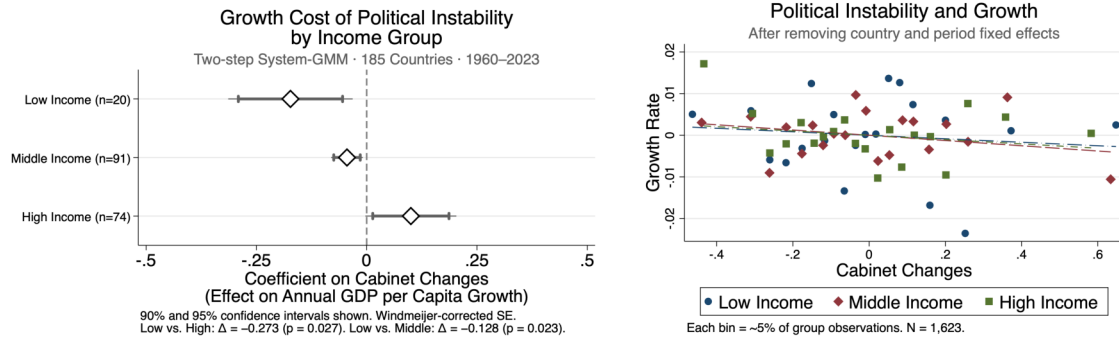
Notes. Dependent variable: log growth of real GDP per capita. Two-step System-GMM with Windmeijer-corrected robust standard errors; t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. AR(2) tests for second-order serial correlation. The borderline Hansen $p = 0.052$ in Column (1) is supported by Difference-in-Hansen subset tests ($p > 0.10$ for all subsets). Hansen $p = 1.000$ in Column (6) indicates the test has near-zero power, most likely due to instrument proliferation or near-collinearity among the instrument columns (Roodman 2009); the over-identification test is uninformative for this column. Interaction terms $PI_{it} \cdot D_i^k$ are treated as strictly exogenous conditional on the instrumented base variable; see Section 3.4.2 for discussion of this limitation. Subsample GMM estimates confirming the directional gradient are reported in Table 8.

Table 6. Robustness Checks: System-GMM (Five-Year Panel)

	(1) Base	(2) Lag $\uparrow$	(3) Lag $\downarrow$	(4) No coll.	(5) –COVID	(6) –GFC	(7) Pre-05	(8) Min.	(9) Trade
Cabinet changes	–0.014** (–2.27)	–0.013** (–2.51)	–0.014** (–2.07)	–0.015*** (–2.98)	–0.014** (–2.27)	–0.019*** (–2.94)	–0.019*** (–2.94)	–0.010 (–1.28)	–0.010 (–1.23)
Initial GDP p.c.	–0.001 (–0.01)	0.016 (0.31)	0.014 (0.23)	–0.018*** (–3.82)	–0.001 (–0.01)	–0.007 (–0.44)	–0.007 (–0.44)	0.031 (0.51)	0.044 (0.62)
Investment	0.001 (0.11)	–0.000 (–0.14)	–0.000 (–0.07)	0.001*** (4.53)	0.001 (0.11)	0.001 (1.44)	0.001 (1.44)	–0.000 (–0.12)	–0.001 (–0.35)
Population growth	–0.511 (–0.27)	–0.130 (–0.11)	–0.165 (–0.10)	–1.047*** (–5.33)	–0.511 (–0.27)	–0.695 (–1.64)	–0.695 (–1.64)	0.269 (0.20)	0.706 (0.40)
Trade	–0.000 (–0.10)	–0.001 (–0.47)	–0.000 (–0.40)	0.000* (1.81)	–0.000 (–0.10)	–0.000 (–0.78)	–0.000 (–0.78)		–0.001 (–0.55)
Inflation	–0.007 (–0.77)	–0.007 (–0.73)	–0.005 (–0.74)	–0.002 (–0.27)	–0.007 (–0.77)	–0.002 (–0.40)	–0.002 (–0.40)		
Government	–0.001 (–0.51)	–0.001 (–1.09)	–0.001 (–1.07)	–0.000 (–1.35)	–0.001 (–0.51)	–0.000* (–1.84)	–0.000* (–1.84)		
Observations	1,297	1,297	1,297	1,297	1,297	752	752	1,623	1,623
Countries	185	185	185	185	185	177	177	185	185
AR(2) p	0.993	0.755	0.816	0.487	0.993	0.596	0.596	0.690	0.542
Hansen p	0.093	0.073	0.001	0.007	0.093	0.014	0.014	0.740	0.830

Notes. Dependent variable: growth rate of real GDP per capita. Two-step System-GMM with Windmeijer-corrected standard errors; t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. AR(1) p -values are not shown to conserve space; all columns produce AR(1) rejection ($p < 0.05$) as expected. Column (3): Hansen test rejects ($p = 0.001$); interpret with caution. Column (4): Hansen test rejects ($p = 0.007$); instrument validity not maintained. Column (5) reproduces Column (1): the COVID period (period5 = 13) is already excluded from the baseline due to missing inflation data. Columns (6) and (7) yield identical estimates: after applying the inflation control, the drop-GFC and pre-2005 samples reduce to the same effective observations ($N = 752$). Estimator comparison across OLS, FE, Difference-GMM, and System-GMM is reported in Table 7.

B. Figures



(a) System-GMM coefficient estimates by income group (b) Conditional correlation (country and period FE removed)

Figure 1. Income-group heterogeneity in the political instability–growth nexus (1960–2023).

Notes. Panel (a) plots income-group-specific System-GMM coefficients on cabinet changes with 90% and 95% Windmeijer-corrected confidence intervals ($N = 1,623$; 185 countries). Panel (b) plots binscatter residuals after removing country and period fixed effects; each bin represents approximately 5% of group observations.

Table 7. Estimator Comparison: OLS, FE, Difference-GMM, and System-GMM

	(1) OLS	(2) FE	(3) Diff-GMM	(4) Sys-GMM
Cabinet changes	−0.0078** (−2.43)	−0.0034 (−0.98)	−0.0121* (−1.88)	−0.0140** (−2.27)
Initial GDP p.c. (log)	−0.0081*** (−5.37)	−0.0496*** (−7.02)	0.0145 (0.25)	−0.0006 (−0.01)
Investment (% GDP)	0.0009*** (5.85)	0.0009*** (3.83)	0.0007** (2.03)	0.0005 (0.11)
Population growth	−0.7208*** (−6.58)	−0.1696 (−0.80)	−0.0629 (−0.17)	−0.5108 (−0.27)
Trade (% GDP)	0.0001 (0.69)	0.0001 (0.74)	−0.0003 (−1.39)	−0.0001 (−0.10)
Inflation (log)	−0.0089 (−1.52)	0.0044 (0.81)	−0.0059 (−0.66)	−0.0072 (−0.77)
Government (% GDP)	−0.0002* (−1.90)	−0.0003 (−1.26)	−0.0012*** (−3.10)	−0.0006 (−0.51)
Observations	1,297	1,297	1,111	1,297
Countries	—	185	184	185
AR(2) p	—	—	0.725	0.993
Hansen p	—	—	0.629	0.093

Notes. Dependent variable: log growth of real GDP per capita; extended sample 1960–2023. Column (1): pooled OLS with period fixed effects and heteroskedasticity-robust standard errors clustered by country. Column (2): within-group FE with period fixed effects, clustered standard errors. Column (3): two-step Difference-GMM with Windmeijer-corrected standard errors; lag structure $\ln y(2,3)$, $\text{polit11}(1,2)$, collapsed. Column (4): two-step System-GMM (same lag structure); reproduces the baseline from Table 6. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. AR(1) is not applicable for OLS and FE; values for Columns (3) and (4) should be retrieved from Stata. The cabinet changes coefficient is negative across all four estimators. The FE estimate loses significance at conventional levels, consistent with attenuation from the within-group transformation in a long panel ($T = 13$), but the point estimate remains negative and directionally consistent with the GMM results.

Table 8. Subsample GMM: Cabinet Changes and Growth by Income Group (1960–2023)

	(1) Low Income	(2) Middle Income	(3) High Income
Cabinet changes	−0.0143 (−1.11)	−0.0211*** (−3.15)	0.0094* (1.81)
Initial GDP p.c. (log)	0.0230* (1.97)	0.0147* (1.81)	−0.0213 (−0.85)
Investment (% GDP)	0.0010* (2.06)	0.0007** (2.42)	0.0009** (2.21)
Population growth	1.5247*** (2.87)	0.1432 (0.34)	−3.2147** (−2.46)
Government (% GDP)	−0.0004 (−0.89)	−0.0005*** (−3.84)	−0.0016* (−1.80)
Observations	255	686	197
Countries	26	69	19
AR(2) p	0.425	0.987	0.583
Hansen p	0.229	0.288	0.676

Notes. Each column estimates equation (5) on the subsample defined by the income group classifier in equation (6). Two-step System-GMM with Windmeijer-corrected standard errors; lag structure $\ln y(2,3)$, $\text{polit11}(1,2)$, collapsed. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Controls: investment, population growth, government consumption, and period dummies. The directional gradient—negative in low- and middle-income countries, positive in high-income—is consistent with the interaction-model estimates in Table 5. The low-income coefficient is statistically insignificant, likely reflecting limited instrument power with only 26 country groups. All three subsamples pass the Hansen test ($p > 0.20$) and the AR(2) test ($p > 0.40$). Country counts (Low: 26, Middle: 69, High: 19) differ from the group sizes reported in Section 3.4.2 (Low: 20, Middle: 91, High: 74); this discrepancy arises because the subsample estimations use only observations for which the full instrument matrix $\ln y(2,3)$ is non-missing, reducing effective coverage relative to the interaction model. The exact mapping should be verified in the Stata do-file and reconciled in a subsequent revision.

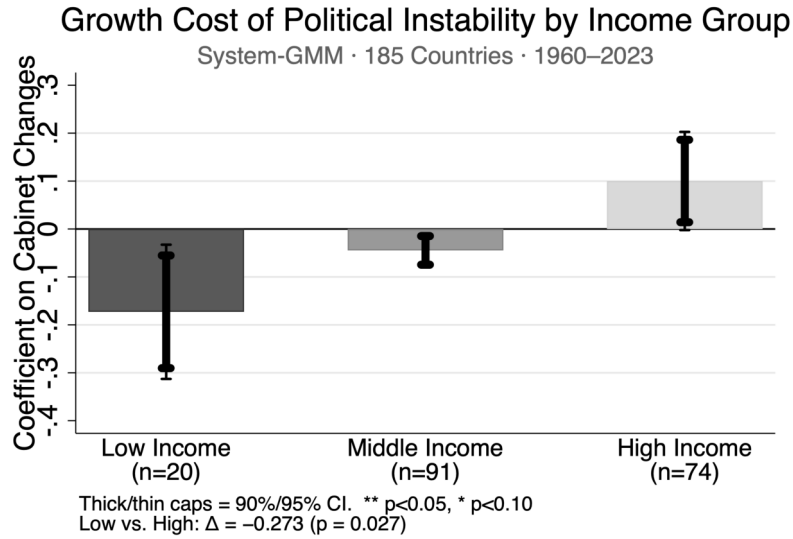
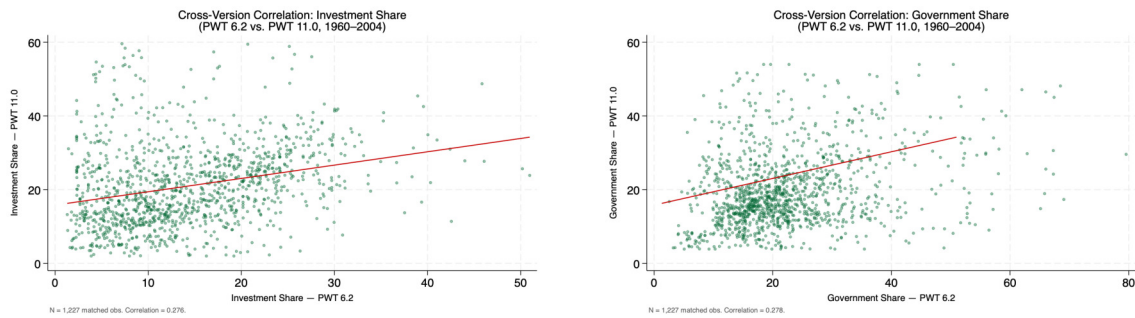


Figure 2. Growth cost of political instability by income group: bar chart representation.
Notes. Bar heights equal the System-GMM point estimates from Column (1) of Table 5. Thick and thin error bars denote 90% and 95% Windmeijer-corrected confidence intervals, respectively. ** $p < 0.05$, * $p < 0.10$.



(a) Investment share ($r = 0.276$)

(b) Government share ($r = 0.278$)

Figure 3. Cross-version correlation: PWT 6.2 vs. PWT 11.0 (1960–2004).

Notes. Each observation is a matched country–period cell from the five-year panel ($N = 1,227$). Red lines are OLS fits.

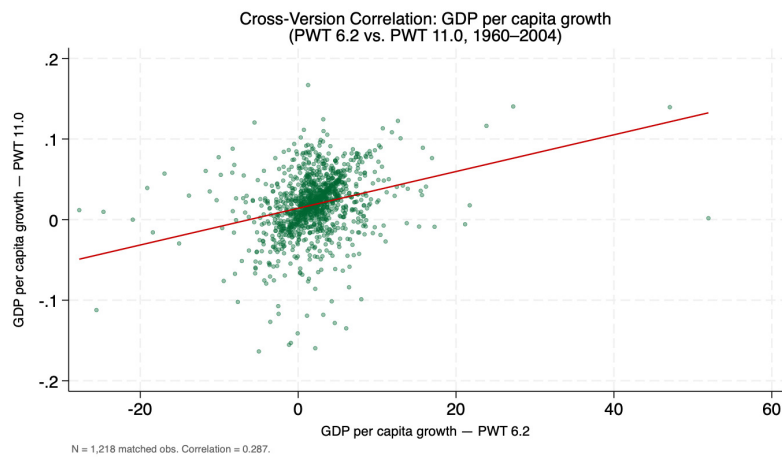


Figure 4. Cross-version correlation: GDP per capita growth, PWT 6.2 vs. PWT 11.0 (1960–2004).

Notes. Each observation is a matched country–period cell from the five-year panel ($N = 1,218$). Red line is OLS fit. The cross-version correlation of $r = 0.287$ is similarly low as those for investment and government shares (see Figure 3), consistent with the broader methodological revisions across PWT versions (Inklaar and Rao 2017).